



University of
Salford
MANCHESTER

Solution of Partial Differential Equations by Finite-Difference Approximations Aerodynamics M1 – Assessment

Report

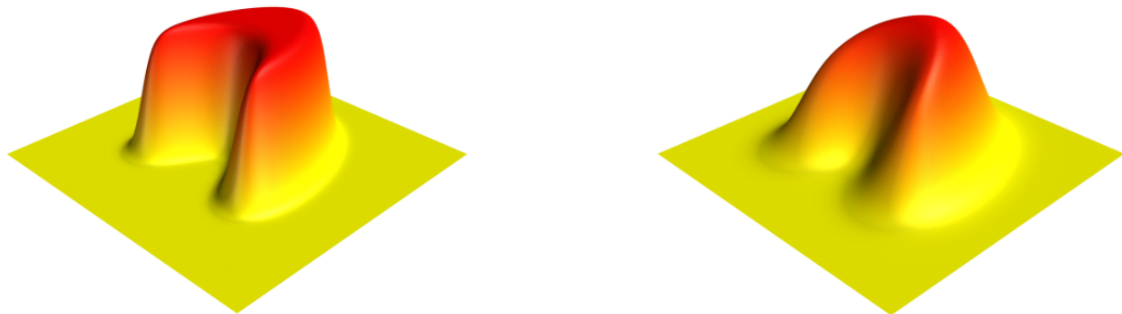


Figure 1: Visualization of an initial profile (left) and its evolution after diffusion (right).

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Abstract

In this assessment, the main objective is to evaluate the numerical solution of linear advection and diffusion equations using finite difference methods, specifically the Lax-Wendroff and Euler methods. These methods often help in studying the stability and consistency of numerical algorithms applied to complex physical models. We will analyze the effects of these parameters using an implementation in Excel, Python, or MATLAB, but in our case, we will use Excel.

We will analyze stability and consistency using Von Neumann methods and Taylor expansions. We will obtain results showing that both methods are stable and converge to suitable solutions as long as we respect the limits established by the instructions. Therefore, they corroborate the validity of Lax's Equivalence Theorem, which states that stable and consistent finite difference schemes will converge to a unique solution of the Partial Differential Equation (PDE)

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1 Introduction

In the mathematical world, analytical models are useful for most physical and engineering problems, but there are many sectors where, when we want to go further, we are very limited by calculation. That is why numerical models are the key for more complex models. For example, if we take the Navier-Stokes or Three-Body Problem (Newtonian Gravitation), which have no analytical solution for the real world, we will use finite difference approximations to discretize these equations and obtain close solutions more efficiently.

In this document, we will discuss two well-known cases of numerical solutions: the linear advection equation and the diffusion equation. These serve to test and validate numerical schemes before applying them to more complex systems and attempting to go further in the calculations and development of our system.

And our main objective here will be to analyze the stability and consistency of these equations by also visualizing the graphs that interpret the data at higher orders to validate the coherence. What is truly interesting is to search and investigate when these approaches can be kept as stable and consistent as possible in order to understand and be able to interpret the most significant results.

2 Theoretical Background

This set of linear advection and diffusion equations is especially straightforward to testing and develop numerical algorithms, as it encapsulates fundamental aspects of more intricate physical equations. The linear advection equation describes the transport of some material (e.g. heat, particles) at constant velocity through a medium. The underlying behavior is crucial in some applications, such as meteorology or fluid dynamics, where it is important to accurately follow motion and fluxes.

$$\frac{\partial U}{\partial t} + A \frac{\partial U}{\partial x} = 0$$

Equation 1 : Linear advection equation

Let U denote the concentration of the species under consideration (in $kg.m^3$), which may also represent density or temperature in some models; nevertheless, we are examining the general case here. Additionally, let t represent time (in s), x signify location (in m), and A denote the advection velocity in the x-direction (in $m.s^{-1}$).

$$\frac{\partial U}{\partial t} = \alpha \frac{\partial^2 U}{\partial x^2}$$

Equation 2 : Diffusion equation

Where α is the (constant) diffusion coefficient.

Before applying finite-difference techniques to more complicated systems, their stability, accuracy, and computational efficiency can be assessed on these simpler models. As a

result, broad numerical techniques utilized in the applied sciences and engineering can be developed and tested using the equations of advection and diffusion.

2.1 Input and Computed Parameters

The numerical analysis is based on the parameters defined in the statement, summarized in Table 1. The spatial domain is $0 \leq x \leq 1$, discretized into 20 equal intervals ($\Delta x = 0.05$).

Table 1: Summary of Problem Parameters and Numerical Schemes

Characteristic	Advection Equation	Diffusion Equation
Equation (PDE)	$\frac{\partial U}{\partial t} + A \frac{\partial U}{\partial x} = 0$	$\frac{\partial U}{\partial t} = \alpha \frac{\partial^2 U}{\partial x^2}$
Physical Constant	$A = 0.5$	$\alpha = 10^{-3}$
Numerical Scheme	Lax-Wendroff	Euler (FTCS)
Dimensionless Parameter	$v = \frac{A \Delta t}{\Delta x}$	$\sigma = \frac{\alpha \Delta t}{\Delta x^2}$
Values to Test (v/σ)	0.25, 0.50, 1.00, 1.25	0.10, 0.25, 0.50, 1.00

2.2 Finite Difference Schemes

The finite difference equations provided in the statement are:

2.2.1 Lax-Wendroff Scheme

$$U_i^{n+1} = U_i^n - \frac{v}{2}(U_{i+1}^n - U_{i-1}^n) + \frac{v^2}{2}(U_{i+1}^n - 2U_i^n + U_{i-1}^n)$$

2.2.2 Euler Scheme (FTCS)

$$U_i^{n+1} = U_i^n + \sigma(U_{i+1}^n - 2U_i^n + U_{i-1}^n)$$

2.3 Excel Implementation and Example Tables

To carry out this numerical implementation we have chosen Excel because this is the model that allows us to implement finite difference models in a more visual way. It is possible to see all the parameters and stages simultaneously, which facilitates tracking and modifying the finite difference schemes and verifying our work, and allows us to handle many more data and interpret them more easily visually.

We will start first with the solution of the linear advection equation using the Lax-Wendroff scheme. First, we will handle the input data with the initial and final positions, x_0 and x_{end} , and the advection velocity $A = 0.5$. We will use this data to initialise the problem.

It is also important to calculate the discretisation steps using the project domain area, and in this case we will focus on determining the temporal and spatial steps (Δt , Δx). Using the values we obtain as fixed ν and the relation:

Data input	
x_0	0
x_{end}	1
A	0.5
ν	0.25
Num points	20

Calculated Parameters	
dx	0.05
dt	0.025

Table 2: Input and Computed Parameters

$$\nu = A \frac{\Delta t}{\Delta x} \quad (1)$$

Okay, and for the setup we will create a row for the spatial part and another for the temporal area which we will express with n , and our goal will be to go up to 15. Then, once created, we will have to set the initial conditions and the boundaries based on the project constraints.

Advection (Lax-Wendroff): The Excel formula we used was:

$$=F18 - (\$F\$6 * 0,5) * (F19 - F17) + ((\$F\$6^2) / 2) * (F19 - 2 * F18 + F17)$$

Where: F18 is U_i^n , $\$F\6 is the value of the parameter ν , F19 is U_{i+1}^n , and F17 is U_{i-1}^n . Shows the complete data set for the stable case $\nu = 0.25$. The other cases ($\nu = 0.50, 1.00, 1.25$) were generated in the same way, simply by changing the time step Δt .

		t (s)															
		0	0,025	0,05	0,075	0,1	0,125	0,15	0,175	0,2	0,225	0,25	0,275	0,3	0,325	0,35	0,375
		n															
x	i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
		0	0	10	10	10	10	10	10	10	10	10	10	10	10	10	10
		0,05	1	10	10	10	10	10	10	10	10	10	10	10	10	10	10
		0,1	2	10	10	10	10	10	10	10	10	10	10	10	10	10	10
		0,15	3	0	15,625	3,1738281	4,776001	6,3145161	7,7407807	9,0440651	10,10231857	10,8661827	11,65245271	12,1046797	12,34173165	12,39018887	12,27053957
		0,2	4	0	0	0,2441406	0,7247925	1,4221668	2,3053547	3,3374527	4,472025103	5,661745297	6,858058489	8,014320332	9,088080676	10,04300577	10,85035894
		0,25	5	0	0	0	0,036147	0,143016	0,3613532	0,6363674	1,0649805	1,7485327	2,51555871	3,372789394	4,324984538	5,343210534	6,39470358
		0,3	6	0	0	0	0	0,0053605	0,028971	0,0834407	0,18552628	0,35441472	0,606236037	0,95138685	1,402201408	1,960276778	2,62437401
		0,35	7	0	0	0	0	0	0,0009313	0,0053842	0,0180766	0,045395863	0,098308117	0,185887467	0,320663441	0,514795303	0,773653393
		0,4	8	0	0	0	0	0	0,0001455	0,000977707	0,003738165	0,010675016	0,025298444	0,052540453	0,098784074	0,17174061	0,280168553
		0,45	9	0	0	0	0	0	0	2,27374E-05	0,000746958	0,002365382	0,006156807	0,013934081	0,028363857	0,053036437	0,092833009
		0,5	10	0	0	0	0	0	0	3,55271E-06	3,05311E-05	0,000145263	0,000505238	0,001433125	0,003510331	0,007633193	0,01542317
		0,55	11	0	0	0	0	0	0	0	5,51162E-07	5,23932E-06	2,79529E-05	0,000194782	0,000321684	0,000849173	0,00193856
		0,6	12	0	0	0	0	0	0	0	0	8,67362E-08	9,08019E-07	5,17071E-06	2,12068E-05	7,00551E-05	0,00019781
		0,65	13	0	0	0	0	0	0	0	0	0	1,35525E-08	1,54584E-07	3,52847E-07	4,20422E-06	1,48713E-05
		0,7	14	0	0	0	0	0	0	0	0	0	0	2,11758E-09	2,61389E-08	1,73325E-07	8,1839E-07
		0,75	15	0	0	0	0	0	0	0	0	0	0	0	3,30872E-10	4,3944E-09	3,1197E-08
		0,8	16	0	0	0	0	0	0	0	0	0	0	0	0	5,16369E-11	7,35023E-10
		0,85	17	0	0	0	0	0	0	0	0	0	0	0	0	0	8,07794E-12
		0,9	18	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		0,95	19	0	0	0	0	0	0	0	0	0	0	0	0	0	0
		1	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 2: Complete Excel Data for Advection (Lax-Wendroff) – Case $\nu = 0.25$ (Stable)

Diffusion (Euler): Similarly, the formula for diffusion was:

$$=F18 + \sigma^2(F19-2*F18+F17)$$

Where: $F18$ is U_i^n , σ^2 is the value of the parameter σ , and $(F19-2*F18+F17)$ is the approximation of the second derivative. Shows the complete data set for the stable case $\sigma = 0.10$.

		t(s)																		
Condition aux limites		0	0,025	0,275	0,525	0,775	1,025	1,275	1,525	1,775	2,025	2,275	2,525	2,775	3,025	3,275	3,525			
Condition initiales		n																		
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15			
x	i	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0			
		0,05	1	0	0	0	0	0	0	0	0	0,0000001	7,3E-07	2,97E-06	8,889E-06	2,183E-05	4,662E-05	8,961E-05	0,0001586	
		0,1	2	0	0	0	0	0	0	0	0,000001	0,0000065	2,386E-05	6,513E-05	0,0001472	0,0002915	0,0005231	0,0008994	0,0013594	
		0,15	3	0	0	0	0	0	0	0,000001	0,0000057	0,0001865	0,0004597	0,0009483	0,0017287	0,0028767	0,0044628	0,0065492	0,0091877	
		0,2	4	0	0	0	0	0,00001	0,000049	0,0003084	0,0007816	0,0016359	0,0031228	0,0052666	0,0086142	0,0132262	0,0193262	0,0265815	0,0349282	
		0,25	5	0	0	0	0	0,0001	0,00041	0,001015	0,001963	0,0032842	0,0049646	0,00698657	0,00931787	0,01192064	0,01475465	0,01777988	0,0205915	
		0,3	6	0	0	0	0,001	0,0033	0,00666	0,011494	0,0199708	0,0206332	0,0257014	0,0332253	0,04318443	0,05004912	0,05683496	0,06348097	0,0694279	
		0,35	7	0	0	0,1	0,25	0,421	0,5965	0,76741	0,929005	1,079106	1,2169957	1,3427744	1,4569791	1,5603603	1,653753	1,7380032	1,8193283	
		0,4	8	0	1	1,7	2,2	2,564	2,8335	3,03583	3,189312	3,306476	3,396077	3,4643703	3,5159262	3,5541563	3,5816587	3,6004479	3,6211014	
		0,45	9	10	9	8,3	7,79	7,402	7,0938	6,83907	6,621259	6,4298558	6,2580913	6,1015252	5,9571745	5,8229766	5,6974559	5,5795173	5,4681371	
		0,5	10	10	10	9,8	9,5	9,158	8,8068	8,4642	8,139174	7,835591	7,554444	7,2951734	7,0564438	6,8365899	6,6338673	6,446585	6,2731715	
		0,55	11	10	9	8,3	7,79	7,402	7,0938	6,83907	6,621259	6,4298558	6,2580913	6,1015252	5,9571745	5,8229766	5,6974559	5,5795173	5,4681371	
		0,6	12	0	1	1,7	2,2	2,564	2,8335	3,03583	3,189312	3,306476	3,396077	3,4643703	3,5159262	3,5541563	3,5816587	3,6004479	3,6211014	
		0,65	13	0	0,1	0,25	0,421	0,5965	0,76741	0,929005	1,079106	1,2169957	1,3427744	1,4569791	1,5603603	1,653753	1,7380032	1,8193283	1,9132928	
		0,7	14	0	0	0	0	0,001	0,0033	0,00666	0,011494	0,0199708	0,0206332	0,0257014	0,0332253	0,04318443	0,05004912	0,05683496	0,06348097	0,0694279
		0,75	15	0	0	0	0	0	0,0001	0,00041	0,001015	0,001963	0,0032842	0,0049646	0,00698657	0,00931787	0,01192064	0,01475465	0,01777988	0,0193282
		0,8	16	0	0	0	0	0	0	0,0001	0,00049	0,001408	0,0030984	0,0057816	0,0096359	0,0147901	0,0213228	0,0292666	0,0386142	0,0495282
		0,85	17	0	0	0	0	0	0	0	0,00001	0,0000057	0,0001865	0,0004597	0,0009483	0,0017287	0,0028767	0,0044628	0,0065492	0,0091877
		0,9	18	0	0	0	0	0	0	0	0,000001	0,0000065	2,386E-05	6,513E-05	0,0001472	0,0002915	0,0005231	0,0008994	0,0013594	
		0,95	19	0	0	0	0	0	0	0	0	0,0000001	7,3E-07	2,97E-06	8,889E-06	2,183E-05	4,662E-05	8,961E-05	0,0001586	
		1	20	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	

Figure 3: T2: Complete Excel Data for Diffusion (Euler) – Case $\sigma = 0.10$ (Stable)

3 Analysis of Schemes

To thoroughly examine the reliability of our numerical solutions, we implemented consistency and stability assessments. These tests validate whether the chosen methods reliably estimate the transport and diffusion equations without introducing significant errors or volatility.

3.1 Consistency Check

The accuracy of a numerical method in approximating a differential equation is determined by its consistency, which is evaluated when the discretization steps (Δx and Δt) approach zero. A consistent approach assures that the numerical solution will converge to the correct solution of the equation as the mesh is adjusted. This check is shown in detail below.

For the Lax–Wendroff finite-difference scheme, after simplification we obtain:

$$U_i^{n+1} = U_i^n + \Delta t \frac{\partial U}{\partial t} + \frac{\Delta t^2}{2} \frac{\partial^2 U}{\partial t^2} + O(\Delta t^3)$$

If $\Delta x \rightarrow 0$ and $\Delta t \rightarrow 0$, we get:

$$U_i^n - \nu \Delta x \frac{\partial U}{\partial x} + \frac{\nu \Delta x^3}{6} \frac{\partial^3 U}{\partial x^3} + O(\Delta x^4)$$

which leads to:

$$U_i^n \approx U_i^n + \frac{\nu^2 \Delta x^2}{2} \frac{\partial^2 U}{\partial x^2} + O(\Delta x^4)$$

Thus, the scheme is consistent with the linear advection equation.

IV.a.ii) Consistency of the Euler scheme

For the explicit Euler scheme, after simplification:

$$U_i^{n+1} = U_i^n + \Delta t \frac{\partial U}{\partial t} + \frac{\Delta t^2}{2} \frac{\partial^2 U}{\partial t^2} + O(\Delta t^3)$$

and also:

$$U_i^{n+1} = U_i^n + \sigma \left(2\Delta x^2 \frac{\partial^2 U}{\partial x^2} \right)$$

When $\Delta x \rightarrow 0$ and $\Delta t \rightarrow 0$, this becomes:

$$U_i^n + O(\Delta t^3) = U_i^n + \sigma O(\Delta x^4)$$

that is:

$$U_i^n \approx U_i^n + O(\Delta x^4)$$

which proves that the Euler scheme is consistent with the diffusion equation.(Appendix 1)

3.2 Stability Check (Von Neumann Analysis)

Now the most important part to verify is how reliable our results will be, therefore we will focus on studying their reliability against the solutions over time, for that we must investigate the stability, and for that we will use the Lax–Wendroff and Euler methods, using the Von Neumann method, which establishes what the stability conditions are with respect to the variables ν and σ .

By stability, we refer to the capacity of our scheme, which, to be precise, is mainly aimed at avoiding the errors grow over the passage of time; this allows that as our calculations go further in time, our problem and our calculations do not deviate into something unrealistic due to small imperfections that grow.

Stability Analysis

In this section we explain how we analyzed and obtained the stability limit using the Von Neumann method for both the Lax–Wendroff scheme (advection) and the explicit Euler scheme (diffusion). First, we recall the conditions required for this method to be applicable, namely: (the partial differential equation and the numerical schemes must be linear, the PDE must have constant coefficients (A and α), and the problem must only depend on two independent variables, in this case x and t). Since these conditions are satisfied, we can conclude that the Von Neumann method can be applied. (Appendix 2)

3.2.1 Lax–Wendroff Stability

After the analysis carried out we obtain this equation as the Lax–Wendroff stability, that is to say the conditions that we must respect so that this can be carried out:

$$\nu = \frac{A\Delta t}{\Delta x} \leq 1$$

3.2.2 Euler Stability

And in this case the same but for our Euler numerical scheme where we see each time the limits in which we must stay so as not to make mistakes or not to have imprecise values:

$$\sigma = \frac{\alpha\Delta t}{\Delta x^2} \leq \frac{1}{2}$$

4 Graphs and Final Discussion

In this section, we will confirm our calculations carried out by hand with the schemes and we will verify the limits and what happens when we approach the limits. We will present the 8 graphs, which we will briefly comment on, and then we will discuss each one and how they behave when they tend towards the limit, and we will verify which values are the closest and which are the farthest from the final solution of the problem.

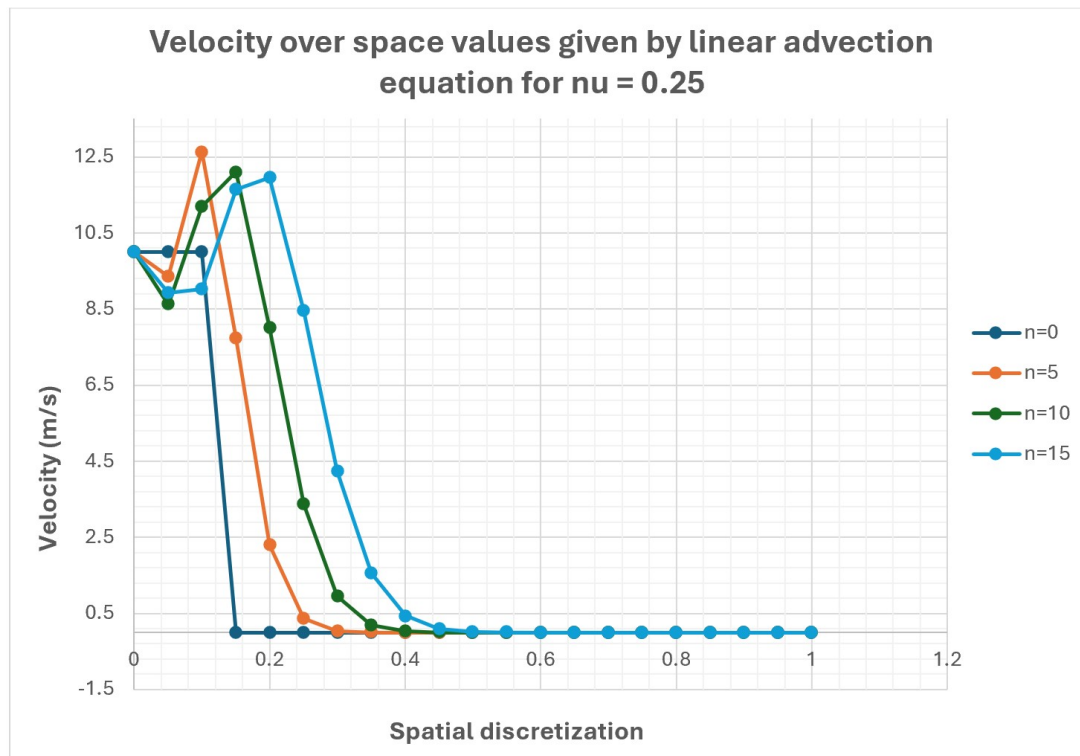


Figure 4: Numerical Advection Results using the Lax–Wendroff Scheme ($\nu = 0.25$).

4.1 Linear Advection Equation (Lax-Wendroff)

4.1.1 Case 1: $\nu = 0.25$ (Stable)

Discussion ($\nu = 0.25$): In the first graph we can see that the structure of the graph when the value is kept at $\nu = 0.50$ gives us a good graph with the shape that we expect, and we observe that it converges quite fast towards zero. This shows that the scheme is stable and that the solution behaves correctly, even if the convergence is relatively fast.

4.1.2 Case 2: $\nu = 0.50$ (Stable)

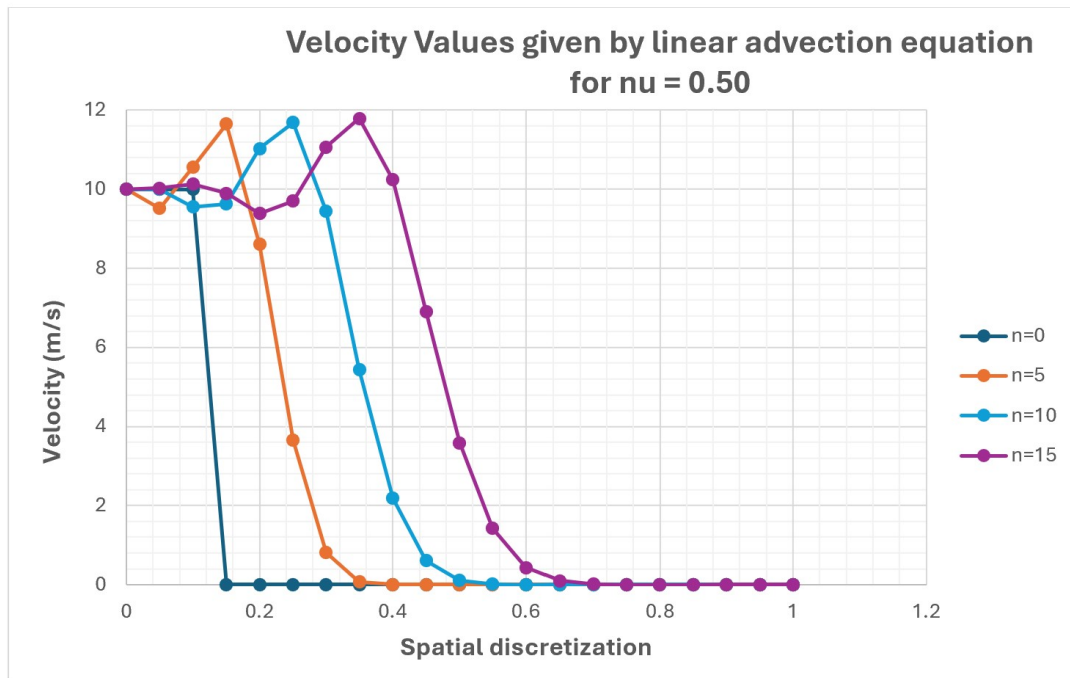


Figure 5: Numerical Advection Results using the Lax–Wendroff Scheme ($\nu = 0.50$).

Discussion ($\nu = 0.50$): In the second one, at $\nu = 0.50$, we observe the same shape of the graph, and we see that it takes a little longer to converge to zero. It does not do it as fast as the previous one, which shows that it is slightly more precise, since the numerical diffusion is smaller and the structure of the solution is better preserved.

4.1.3 Case 3: $\nu = 1.00$ (Stability Limit)

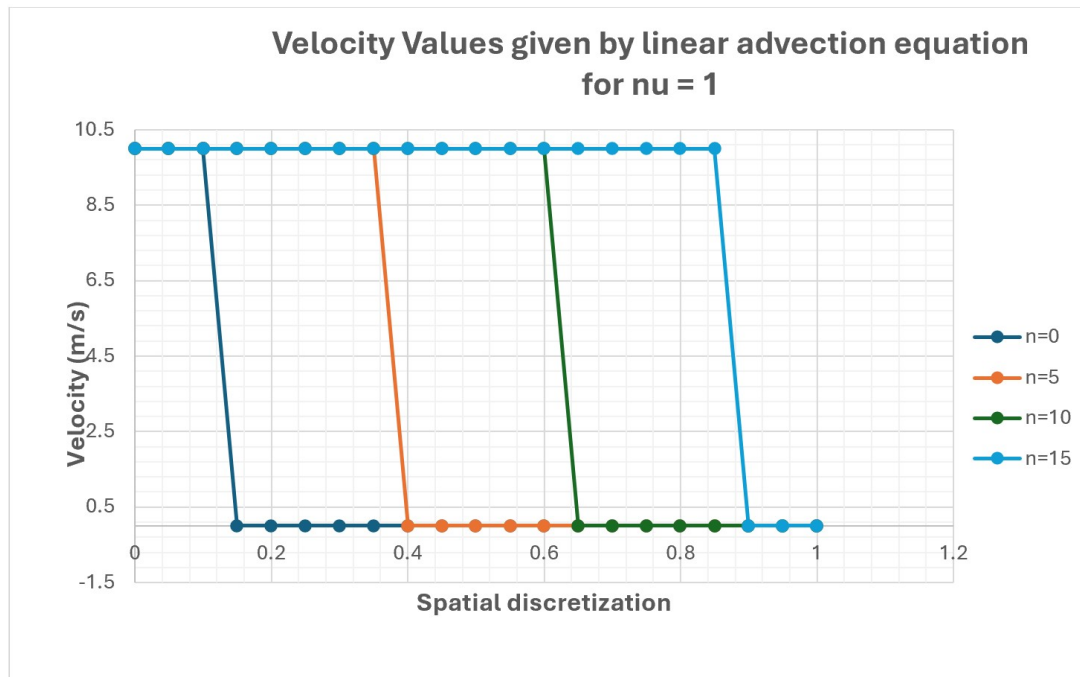


Figure 6: Numerical Advection Results using the Lax–Wendroff Scheme ($\nu = 0.1$).

Discussion ($\nu = 1.00$): The case $\nu = 1.00$. When we place ourselves at the $\nu = 1.00$, we observe that the solution still converges to zero, but in this case the structure of the wave starts to deform and the general tendency of the graph is no longer the same. This can lead us to think that in this case the solution becomes much less precise and starts to move away from the physical reality

4.1.4 Case 4: $\nu = 1.25$ (Unstable)

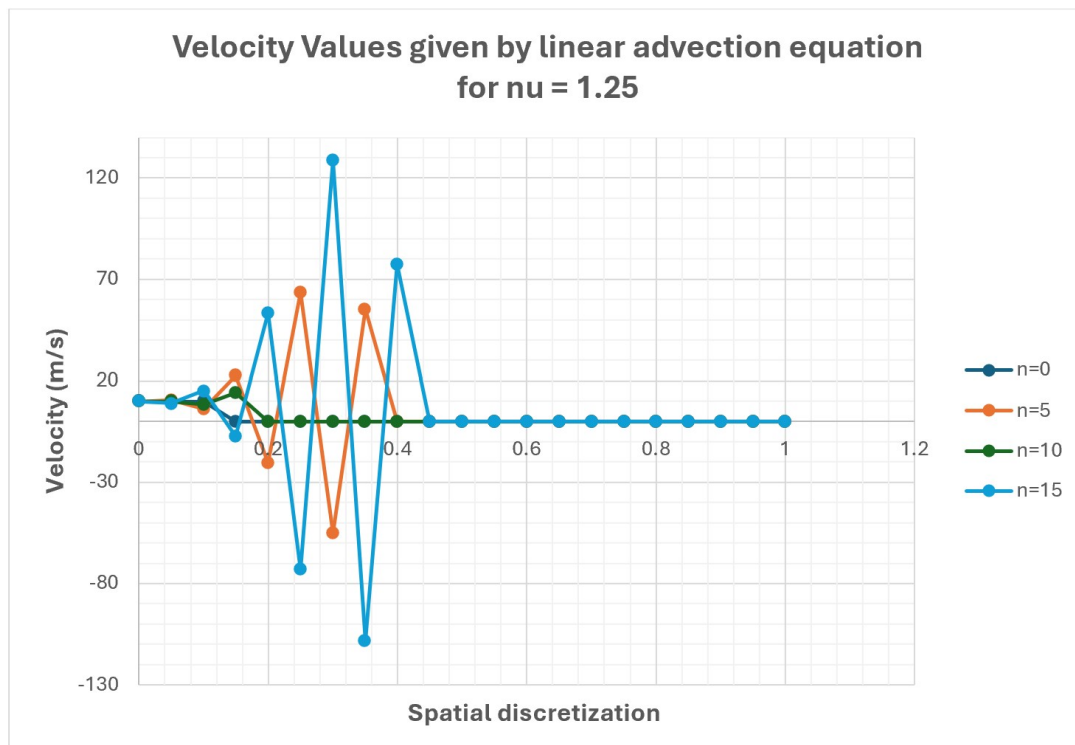


Figure 7: Numerical Advection Results using the Lax–Wendroff Scheme ($\nu = 1.25$) (Unstable).

Discussion ($\nu = 1.25$): Finally, for $\nu = 1.25$, we observe oscillations everywhere, the tendencies are no longer the same and the graphs become very imprecise. This clearly shows that the scheme is unstable for this value and that the numerical solution has no physical meaning anymore.

4.2 Diffusion Equation (Euler Scheme)

4.2.1 Case 5: $\sigma = 0.10$ (Stable)

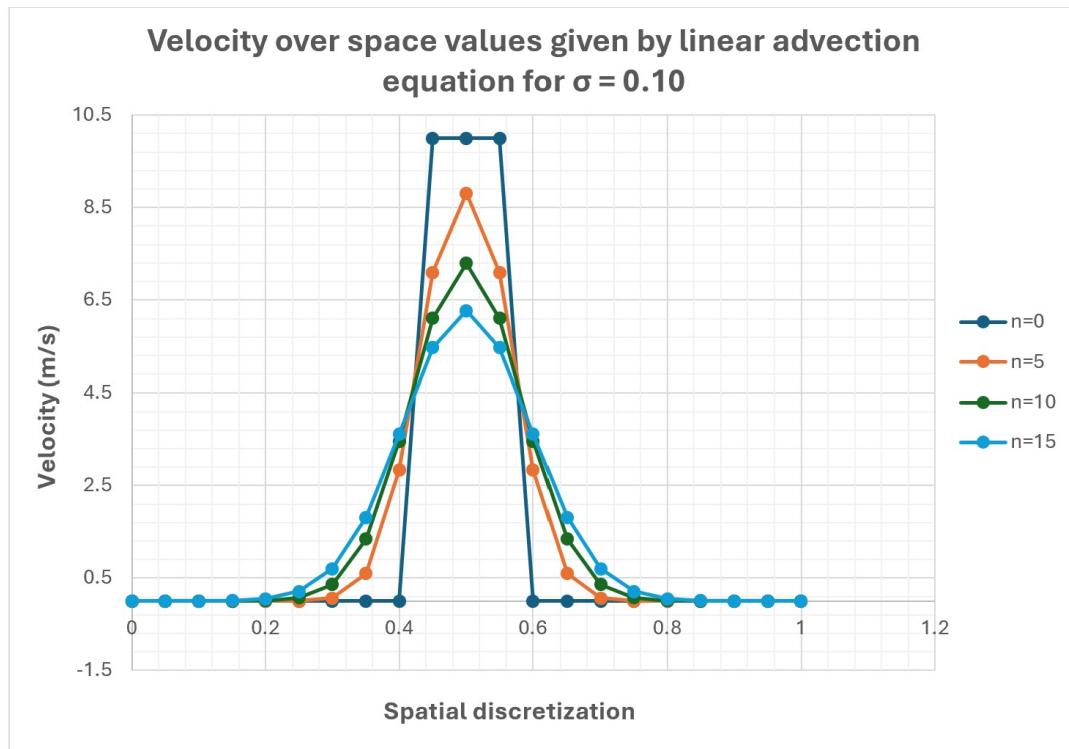


Figure 8: Numerical Diffusion Results using the Euler Method for $\sigma = 0.10$.

Discussion ($\sigma = 0.10$): For $\sigma = 0.10$, we observe that the graph tends to zero quite rapidly and the solution keeps the shape of a smooth curve. This indicates that the scheme is stable and that the diffusion process behaves in a physically correct way, even though the dissipation is relatively fast.

4.2.2 Case 6: $\sigma = 0.25$ (Stable)

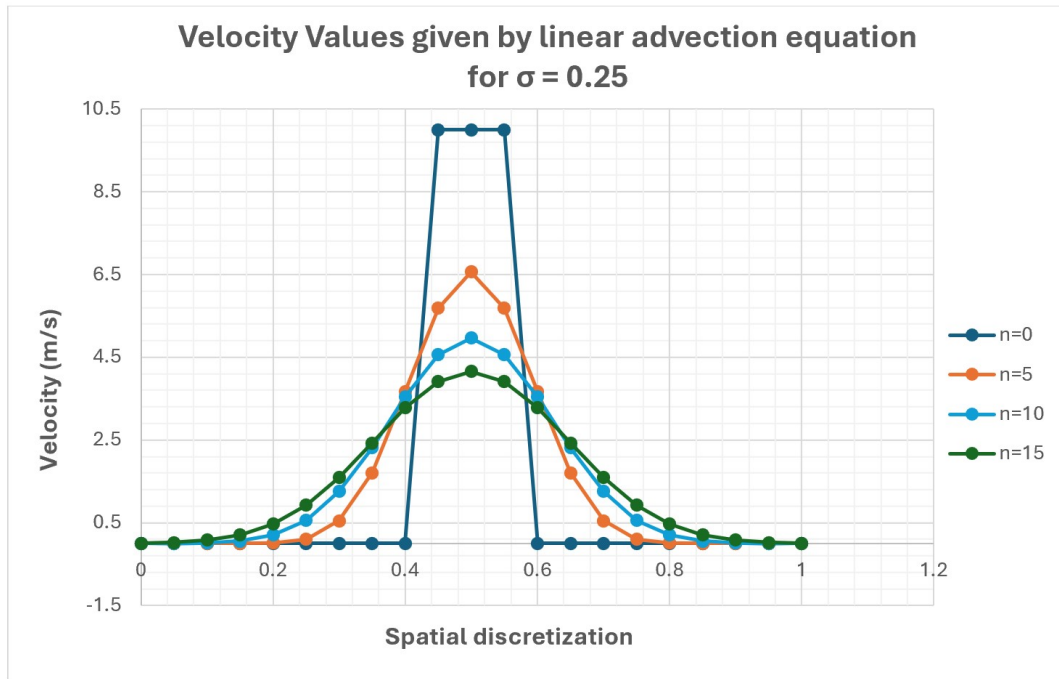


Figure 9: Numerical Diffusion Results using the Euler Method for $\sigma = 0.25$.

Discussion ($\sigma = 0.25$): For $\sigma = 0.25$, the graph follows the same general tendency and remains stable. However, we observe that it takes a longer time to tend towards zero and the solution involves more intermediate values. This shows that the solution is slightly more precise and closer to the real physical solution than in the previous case.

4.2.3 Case 7: $\sigma = 0.50$ (Stability Limit)

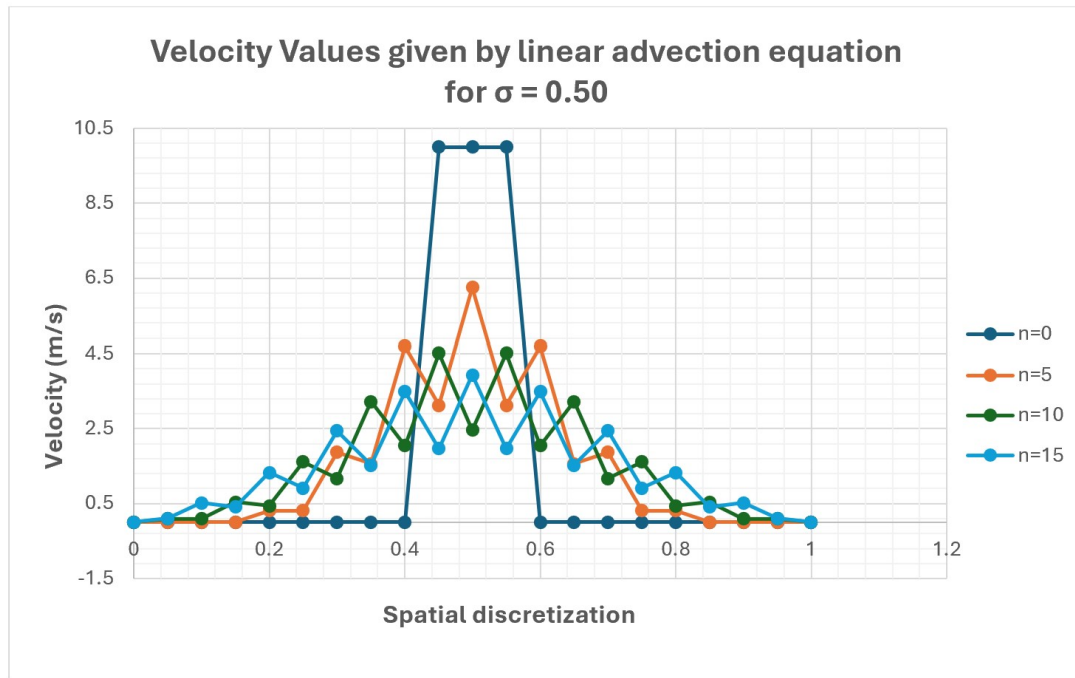


Figure 10: Numerical Diffusion Results using the Euler Method for $\sigma = 0.50$ (Stability Limit).

Discussion ($\sigma = 0.50$): For $\sigma = 0.50$, we are at the stability limit. In this case, the trend of the curve is slightly deformed but it still follows the same overall behavior. Logically, we move further away from the physical reality, but since we are exactly at the stability limit, the solution can still be considered as valid.

4.2.4 Case 8: $\sigma = 1.00$ (Unstable)

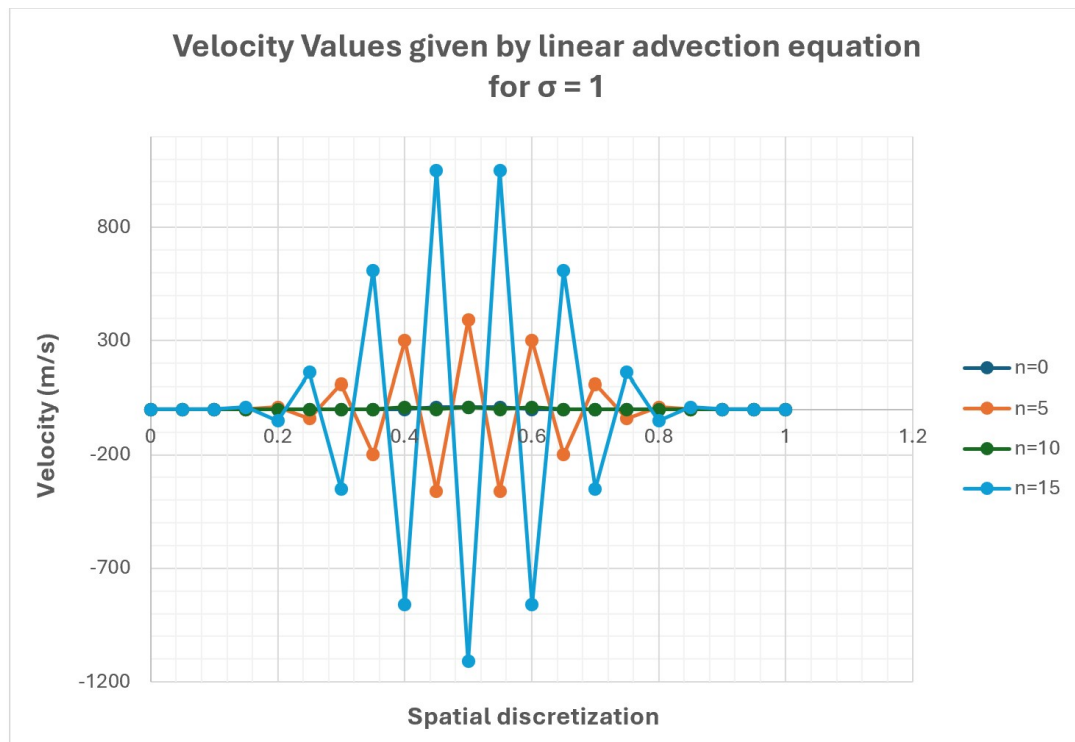


Figure 11: Numerical Diffusion Results using the Euler Method for $\sigma = 1.00$ (Unstable).

Discussion ($\sigma = 1.00$): For the last case, where $\sigma = 1.00$, the scheme becomes unstable. We can see that the curve oscillates in several regions and the numerical solution becomes very imprecise, moving far away from the physical behavior. This confirms that this value of σ leads to numerical instability and unrealistic results.

4.3 Discussion on Accuracy Verification

In general, after the analysis of the graphs, we can talk about the importance of respecting the mathematical conditions that we have applied to our numerical schemes, in this case consistency (that is the approximation when $\Delta t \rightarrow 0$ and $\Delta x \rightarrow 0$) and the stability conditions of our schemes.

We deduce that if one of these two properties fails, we lose physical and mathematical validity and our final results are no longer valid, or depending on how close we are to the limit, they become very unreliable. This is why we see that the Von Neumann study we performed is not only a theoretical calculation, but it has a direct impact on our graphs and also directly on the solution that we obtain.

For example, for the Advection equation using the Lax–Wendroff scheme, when we are in the case:

$$\nu < 1$$

the shape of the curve remains coherent with the physical solution, without producing strong oscillations, conserving the structure and the shape of the solution. On the contrary, when:

$$\nu > 1$$

strong oscillations and non-physical peaks appear, the tendency of the curve is completely lost, and we clearly observe an instability in our values and in the solution. This has the consequence that the solution becomes false or very far from reality, and the scheme no longer represents correctly the physical phenomenon.

The stability condition is:

$$\nu = \frac{A \Delta t}{\Delta x} \leq 1$$

which shows a direct relation between stability and consistency.

The same reasoning applies to the diffusion equation using the explicit Euler scheme. When we are below the stability limit:

$$\sigma = \frac{\alpha \Delta t}{\Delta x^2} \leq \frac{1}{2}$$

that is, in the stable zone, as shown in our graphical analysis, the curve remains smooth and we progressively approach the real solution. However, when we go beyond this limit and we are not stable anymore:

$$\sigma > 0.5$$

we observe strong oscillations everywhere and the numerical values become completely unrealistic, so the solution loses all physical meaning.

This leads us directly to the Lax Equivalence Theorem, which states that a numerical scheme is valid, consistent and convergent if and only if it is stable. Therefore, we can affirm that only when the stability conditions:

$$\nu < 1 \quad \text{and} \quad \sigma < 0.5$$

are respected, the solutions converge towards physically reasonable results. When these conditions are not respected, numerical instabilities appear and completely invalidate the simulation.

The most interesting question is then: what are the best values to choose inside the stable region in order to obtain a compromise between stability, precision and efficiency?

From our results, we observe that when the values are very low, the schemes are very stable but also very diffusive and less precise. On the other hand, when we get too close to the limit, the schemes become less stable and less accurate.

For this reason, we can consider optimal intermediate ranges such as:

$$0.5 < \nu < 0.8 \quad \text{for Lax-Wendroff}$$

and

$$0.3 < \sigma < 0.4 \quad \text{for explicit Euler diffusion}$$

In this region, we keep enough stability while improving the precision of the solution. Moreover, the computation also becomes more efficient. Indeed, we recall the definition:

$$\nu = \frac{A\Delta t}{\Delta x}$$

which shows that when ν increases, Δt also increases. Since the number of iterations N is given by:

$$N = \frac{T_{\text{final}}}{\Delta t}$$

a larger Δt leads to fewer iterations, which makes the simulation faster. At the same time, a larger Δt allows the solution to evolve more smoothly, reducing artificial diffusion and keeping more physical values during the decay towards zero.

This confirms that choosing intermediate values inside the stability range makes the simulation both more precise and more efficient, which is one of the main goals in numerical modeling.

5 Conclusion

In conclusion, we can state that both the Euler and Lax-Wendroff schemes are consistent and stable, in coherence with the computer simulations that we carried out. Therefore, according to the Lax Equivalence Theorem, they converge properly.

Thus, if we stay within the established limits, we can correctly converge towards a real solution of the differential equation.

We can therefore affirm that if the values of ν and σ remain inside the stability limits, namely:

$$\nu \leq 1 \quad \text{and} \quad \sigma \leq \frac{1}{2},$$

then they can be used with confidence and both numerical schemes are valid.

In addition, we can also include the factor of choosing the most coherent values when we are looking for both precision and computational efficiency. In this study, we have estimated a highly performant interval for these parameters, where the schemes remain stable while also being more accurate and faster in terms of computation time.

Appendix A – Truncation Error and Consistency Derivation

(This material has been integrated into the main body of the report, Section 3.1. This appendix may therefore be left empty or removed.)

$$U_i^{n+1} = U_i^n - \frac{\nu}{2}(U_{i+1}^n - U_{i-1}^n) + \frac{\nu^2}{2}(U_{i+1}^n - 2U_i^n + U_{i-1}^n)$$

is consistent with the advection PDE: $\frac{\partial U}{\partial t} + A \frac{\partial U}{\partial x} = 0$.

To this end, we expand each term of the numerical equation in a Taylor series around the point (n, i) :

$$\begin{aligned} U_i^{n+1} &= U_i^n + \Delta t \frac{\partial U}{\partial t} + \frac{\Delta t^2}{2} \frac{\partial^2 U}{\partial t^2} + O(\Delta t^3) \\ U_{i+1}^n &= U_i^n + \Delta x \frac{\partial U}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 U}{\partial x^2} + \frac{\Delta x^3}{6} \frac{\partial^3 U}{\partial x^3} + O(\Delta x^4) \\ U_{i-1}^n &= U_i^n - \Delta x \frac{\partial U}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 U}{\partial x^2} - \frac{\Delta x^3}{6} \frac{\partial^3 U}{\partial x^3} + O(\Delta x^4) \end{aligned}$$

We substitute these expansions into the scheme. The left-hand side (temporal) is:

$$\text{LHS} = U_i^n + \Delta t \frac{\partial U}{\partial t} + \frac{\Delta t^2}{2} \frac{\partial^2 U}{\partial t^2} + O(\Delta t^3)$$

The right-hand side (spatial) simplifies:

$$\begin{aligned} \text{RHS} &= U_i^n - \frac{\nu}{2} \left((U_i^n + \Delta x \frac{\partial U}{\partial x} + \dots) - (U_i^n - \Delta x \frac{\partial U}{\partial x} + \dots) \right) \\ &\quad + \frac{\nu^2}{2} \left((U_i^n + \Delta x \frac{\partial U}{\partial x} + \dots) - 2U_i^n + (U_i^n - \Delta x \frac{\partial U}{\partial x} + \dots) \right) \end{aligned}$$

Grouping the terms from the expansions, the right-hand side becomes:

$$\begin{aligned} \text{RHS} &= U_i^n - \frac{\nu}{2} \left(2\Delta x \frac{\partial U}{\partial x} + \frac{\Delta x^3}{3} \frac{\partial^3 U}{\partial x^3} + \dots \right) + \frac{\nu^2}{2} \left(\Delta x^2 \frac{\partial^2 U}{\partial x^2} + \dots \right) \\ \text{RHS} &= U_i^n - \nu \Delta x \frac{\partial U}{\partial x} + \frac{\nu^2 \Delta x^2}{2} \frac{\partial^2 U}{\partial x^2} - \frac{\nu \Delta x^3}{6} \frac{\partial^3 U}{\partial x^3} + \dots \end{aligned}$$

Now, we equate the LHS and RHS, and substitute $\nu = \frac{A\Delta t}{\Delta x}$ (the CFL number):

$$U_i^n + \Delta t \frac{\partial U}{\partial t} + \frac{\Delta t^2}{2} \frac{\partial^2 U}{\partial t^2} + \dots = U_i^n - (A\Delta t) \frac{\partial U}{\partial x} + \frac{(A\Delta t)^2}{2} \frac{\partial^2 U}{\partial x^2} + \dots$$

From the original PDE, we know that $\frac{\partial U}{\partial t} = -A \frac{\partial U}{\partial x}$, and differentiating again, $\frac{\partial^2 U}{\partial t^2} = A^2 \frac{\partial^2 U}{\partial x^2}$. If we substitute this into the left-hand side:

$$U_i^n + \Delta t \left(-A \frac{\partial U}{\partial x} \right) + \frac{\Delta t^2}{2} \left(A^2 \frac{\partial^2 U}{\partial x^2} \right) + \dots = U_i^n - A\Delta t \frac{\partial U}{\partial x} + \frac{A^2 \Delta t^2}{2} \frac{\partial^2 U}{\partial x^2} + \dots$$

We observe that the zero-order, first-order, and second-order terms coincide exactly. Upon rearranging the equation to isolate the PDE, we find that:

$$\frac{\partial U}{\partial t} + A \frac{\partial U}{\partial x} = O(\Delta t^2, \Delta x^2)$$

The truncation error $O(\Delta t^2, \Delta x^2)$ tends to zero as $\Delta t, \Delta x \rightarrow 0$. Therefore, the scheme is **consistent** and has an accuracy of **second order** in both time and space.

A.0.1 Euler FTCS Scheme (Diffusion)

We repeat the process for the Euler (FTCS) scheme applied to the diffusion equation $\frac{\partial U}{\partial t} = \alpha \frac{\partial^2 U}{\partial x^2}$. The numerical scheme is:

$$U_i^{n+1} = U_i^n + \sigma(U_{i+1}^n - 2U_i^n + U_{i-1}^n)$$

Expanding using Taylor series:

$$\begin{aligned} \text{LHS} &= U_i^n + \Delta t \frac{\partial U}{\partial t} + \frac{\Delta t^2}{2} \frac{\partial^2 U}{\partial t^2} + O(\Delta t^3) \\ \text{RHS} &= U_i^n + \sigma \left((\Delta x^2 \frac{\partial^2 U}{\partial x^2} + \frac{\Delta x^4}{12} \frac{\partial^4 U}{\partial x^4} + \dots) \right) \end{aligned}$$

We substitute $\sigma = \frac{\alpha \Delta t}{\Delta x^2}$ into the right-hand side:

$$\text{RHS} = U_i^n + \left(\frac{\alpha \Delta t}{\Delta x^2} \right) \left(\Delta x^2 \frac{\partial^2 U}{\partial x^2} + O(\Delta x^4) \right) = U_i^n + \alpha \Delta t \frac{\partial^2 U}{\partial x^2} + O(\Delta t \Delta x^2)$$

We equate both sides:

$$U_i^n + \Delta t \frac{\partial U}{\partial t} + O(\Delta t^2) = U_i^n + \alpha \Delta t \frac{\partial^2 U}{\partial x^2} + O(\Delta t \Delta x^2)$$

We subtract U_i^n from both sides and divide by Δt :

$$\frac{\partial U}{\partial t} + O(\Delta t) = \alpha \frac{\partial^2 U}{\partial x^2} + O(\Delta x^2)$$

Reorganizing to show the PDE and the truncation error:

$$\frac{\partial U}{\partial t} - \alpha \frac{\partial^2 U}{\partial x^2} = O(\Delta t, \Delta x^2)$$

The error term $O(\Delta t, \Delta x^2)$ tends to zero as $\Delta t, \Delta x \rightarrow 0$. Therefore, the scheme is **consistent**. However, we note that it is **first order in time** ($O(\Delta t)$) and **second order in space** ($O(\Delta x^2)$).

Appendix B Appendix B – Stability Behaviour of Numerical Waves

Let us consider the linear advection equation:

$$\frac{\partial u}{\partial t} + A \frac{\partial u}{\partial x} = 0$$

The Lax–Wendroff scheme can be written as:

$$u_i^{n+1} = u_i^n - \frac{\nu}{2}(u_{i+1}^n - u_{i-1}^n) + \frac{\nu^2}{2}(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$$

where the Courant number is defined as:

$$\nu = \frac{A\Delta t}{\Delta x}$$

We define the numerical error as:

$$\varepsilon_i^n = U_i^n - u_i^n$$

We assume that the error behaves like a Fourier mode:

$$\varepsilon_i^n = E^n e^{ikx_i}$$

where E^n is the error amplitude at time level n .

By substituting this expression into the numerical scheme, we obtain:

$$G = \frac{E^{n+1}}{E^n} = 1 - i\nu \sin(\theta) + \nu^2(\cos \theta - 1)$$

where $\theta = k\Delta x$.

For the scheme to be stable, the amplification factor must satisfy:

$$|G| \leq 1$$

Considering the worst-case scenario, we finally obtain the condition:

$$\nu \leq 1$$

Therefore, the Lax–Wendroff scheme is stable if:

$$\boxed{\frac{A\Delta t}{\Delta x} \leq 1}$$

—

A1.b) Stability of the Euler Scheme (Diffusion)

We now consider the diffusion equation:

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}$$

The explicit Euler scheme (FTCS) is written as:

$$u_i^{n+1} = u_i^n + \sigma(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$$

where:

$$\sigma = \frac{\alpha \Delta t}{\Delta x^2}$$

The numerical error is defined in the same way:

$$\varepsilon_i^n = U_i^n - u_i^n$$

and we again assume a Fourier form:

$$\varepsilon_i^n = E^n e^{ikx_i}$$

Substituting into the scheme gives:

$$G = 1 - 4\sigma \sin^2\left(\frac{\theta}{2}\right)$$

where $\theta = k\Delta x$.

The stability condition is:

$$|G| \leq 1$$

Since $\sin^2(\theta/2) \geq 0$, we finally obtain:

$$\sigma \leq \frac{1}{2}$$

That is, the explicit Euler scheme for diffusion is stable if:

$$\boxed{\frac{\alpha \Delta t}{\Delta x^2} \leq \frac{1}{2}}$$

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